

1045 ▶ 1145

1066

Halley's comet

The appearance of Halley's comet just before the Norman defeat of the English at the Battle of Hastings was later explained as the cause of the disaster. Having no real explanation for comets, people generally believed they were evil omens.



In the Bayeux Tapestry (c 1080) people point at the comet



Constantine the African lectures at the school of Salerno, Italy

1085

Medical writings spread

Around this date, Constantine the African, a North African Muslim who converted to Christianity, collected Arabic medical manuscripts such as Haly Abbas's *Complete Art of Medicine*. He translated these at the medical school at Salerno in Italy and helped spread Arabic medical knowledge to Europe.

In 1121, philosopher Abu'l-Barakat of Baghdad proposed that the more force applied to an object, the greater its acceleration.



1045

1070

1095



Jia Xian's version of the mathematical pattern called Pascal's triangle

1054

Sighting of Crab Nebula

On July 4, Chinese astronomers noticed a new star so bright it could be seen in daylight. They called it the "guest star." What they saw was a supernova (exploding star), whose collapse, caused by extreme gravity, formed the cloud of debris in outer space that we now call the Crab Nebula.

1085

Translation of Ptolemy's *Almagest*

When Christian Spanish king Alfonso VI captured Toledo, which had been under Islamic rule, the city became a center for the translation of Arabic scientific works into Latin. One of the most important was Ptolemy's great work on ancient astronomy, the *Almagest*.

1050

Pascal's triangle

The Chinese mathematician Jia Xian created a version of the number pattern that today we know as Pascal's triangle, in which each number is the sum of the two numbers above it. This found later use in calculations involving probability.

An image of the fast-rotating Crab Nebula from NASA's Chandra X-ray observatory.



1088

Magnetic compass described

Spoon-shaped, lodestone compasses had been used in China since around 200 BCE. In 1088 CE, the Chinese scholar Shen Kuo gave the first description of a compass with a magnetized needle. He included it in his work *Dream Pool Essays*, which also contained a discussion of fossils. By the early 12th century, Chinese ships were navigating by compass.